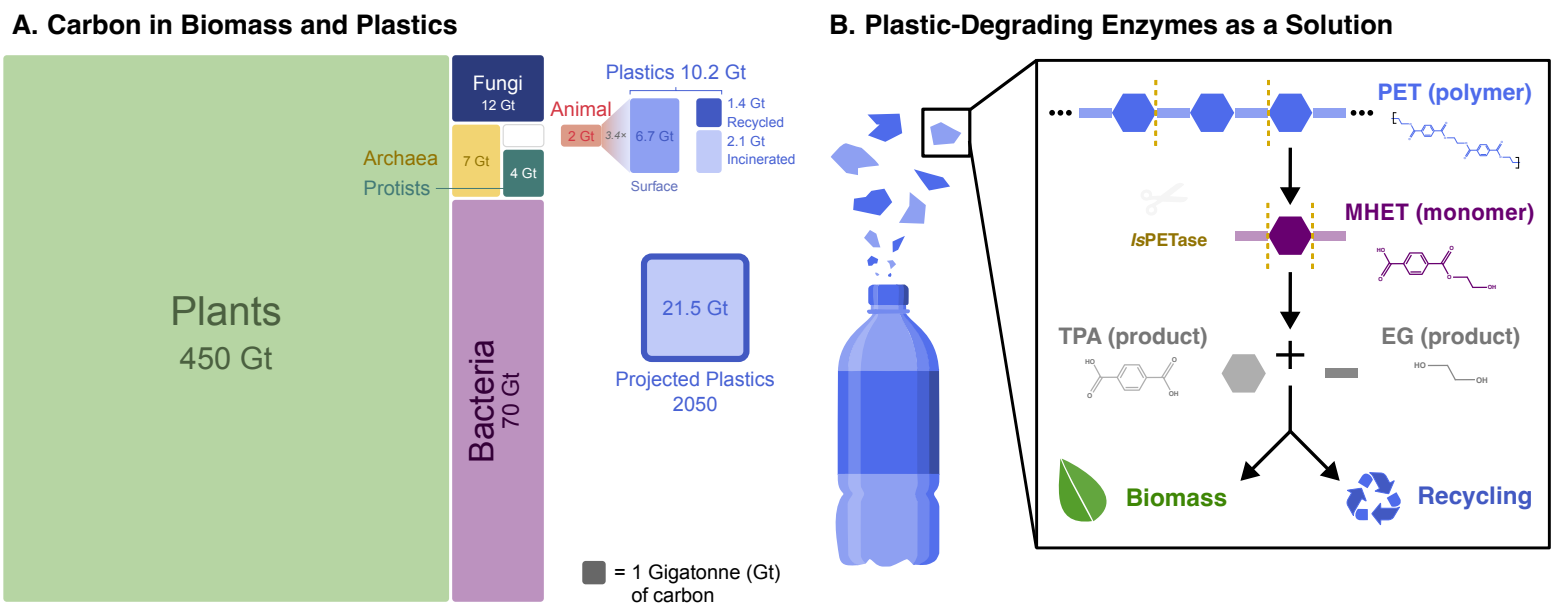

A. Carbon in Biomass and Plastics
Plants
450 Gt
Fungi
12 Gt
Archaea
Protists
7 Gt
4 Gt
Bacteria
70 Gt
Plastics 10.2 Gt
Animal
2 Gt
3.4×
6.7 Gt
Surface
1.4 Gt
Recycled
2.1 Gt
Incinerated
21.5 Gt
Projected Plastics
2050
= 1 Gigatonne (Gt)
of carbon
B. Plastic-Degrading Enzymes as a Solution
PET (polymer)
IsPETase
MHET (monomer)
TPA (product)
EG (product)
Biomass
Recycling

Fig 2A. PETadex-PAZy Composition

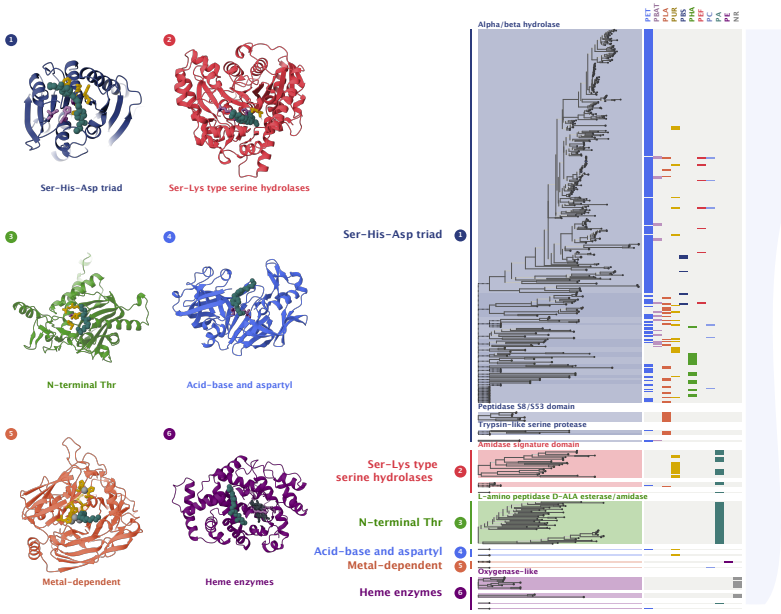


Fig 2B. PETadex Generation

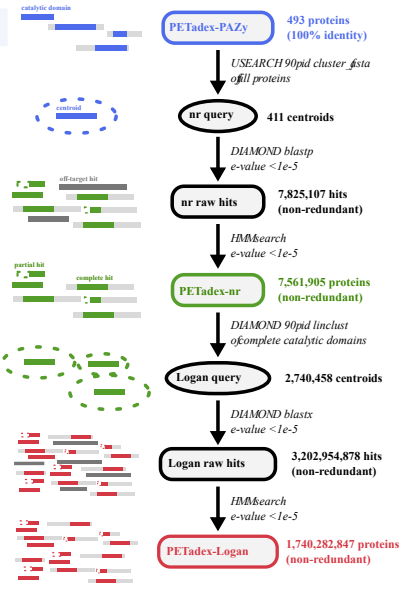


Fig 2C. PETadex Clustering

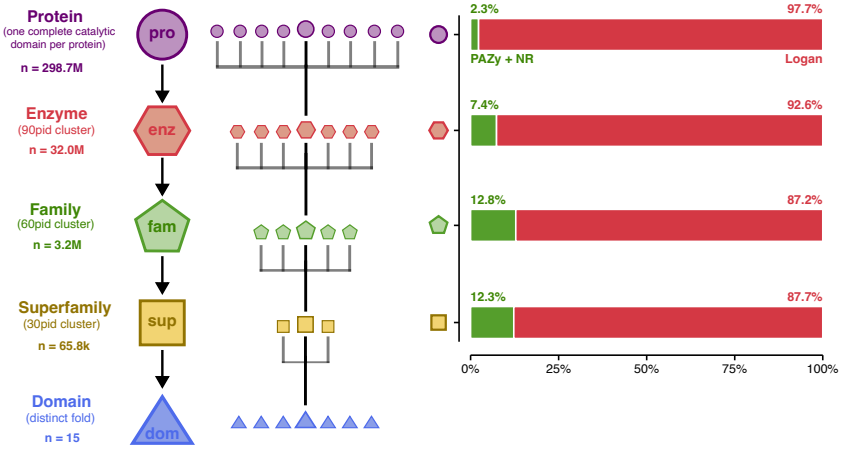
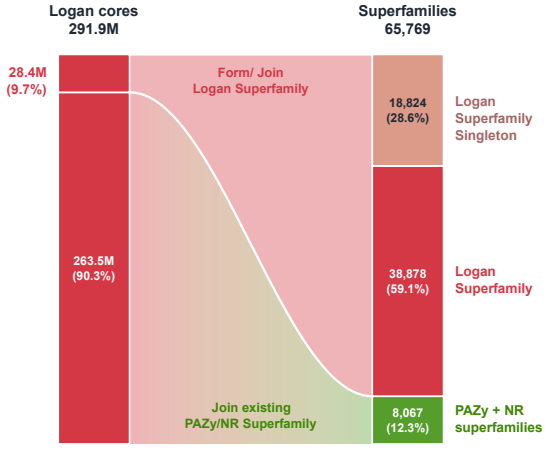
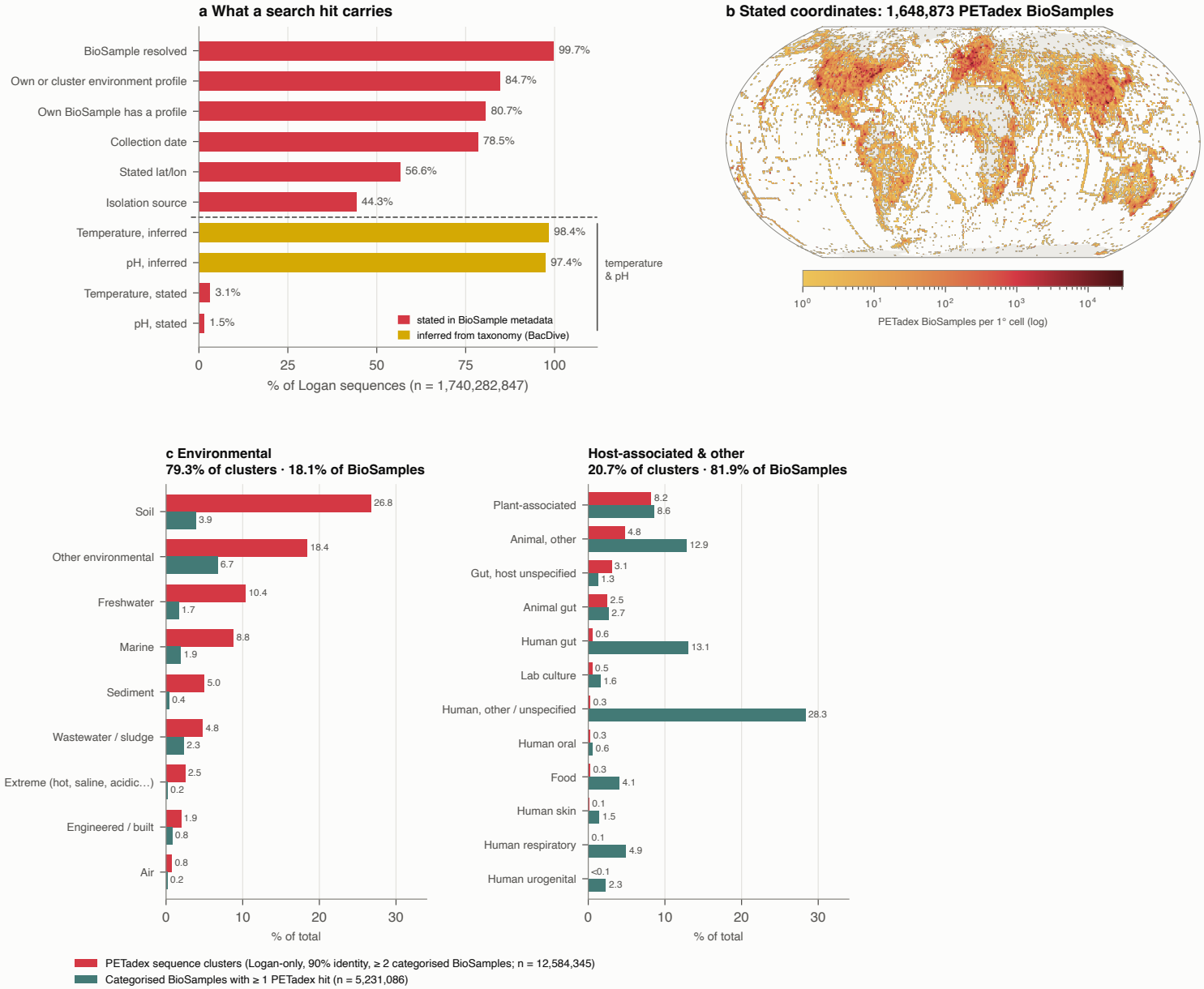


Fig 2D. Logan Contribution




a What a search hit carries
BioSample resolved
Own or cluster environment profile
Own BioSample has a profile
Collection date
Stated lat/lon
Isolation source
Temperature, inferred
pH, inferred
Temperature, stated
pH, stated
99.7%
84.7%
80.7%
78.5%
56.6%
44.3%
98.4%
97.4%
3.1%
1.5%
temperature & pH
stated in BioSample metadata
inferred from taxonomy (BacDive)
0
25
50
75
100
% of Logan sequences (n = 1,740,282,847)
b Stated coordinates: 1,648,873 PETadex BioSamples
10^0
10^1
10^2
10^3
10^4
PETadex BioSamples per 1° cell (log)
c Environmental
79.3% of clusters · 18.1% of BioSamples
Soil
Other environmental
Freshwater
Marine
Sediment
Wastewater / sludge
Extreme (hot, saline, acidic…)
Engineered / built
Air
26.8
3.9
18.4
6.7
10.4
1.7
8.8
1.9
5.0
0.4
4.8
2.3
2.5
0.2
1.9
0.8
0.8
0.2
0
10
20
30
% of total
Host-associated & other
20.7% of clusters · 81.9% of BioSamples
Plant-associated
Animal, other
Gut, host unspecified
Animal gut
Human gut
Lab culture
Human, other / unspecified
Human oral
Food
Human skin
Human respiratory
Human urogenital
8.2
8.6
4.8
12.9
3.1
1.3
2.5
2.7
0.6
13.1
0.5
1.6
0.3
28.3
0.3
0.6
0.3
4.1
0.1
1.5
0.1
4.9
<0.1
2.3
0
10
20
30
% of total
PETadex sequence clusters (Logan-only, 90% identity, ≥ 2 categorised BioSamples; n = 12,584,345)
Categorised BioSamples with ≥ 1 PETadex hit (n = 5,231,086)